

8-b Precision 8-Mb ReRAM Compute-in-Memory Macro Using Direct-Current-Free Time-Domain Readout Scheme for AI Edge Devices

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Abstract—Compute-in-memory (nvCIM) macros based on non-volatile memory make it possible for artificial intelligence (AI) edge devices to perform energy-efficient multiply-and-accumulate (MAC) operations by minimizing the movement of data between the processors and memory. However, nvCIM imposes tradeoffs between energy efficiency, computing latency, and readout accuracy against process variation. To overcome these challenges, this work proposed a nvCIM macro featuring: 1) a direct-current-free time-space-based in-memory computing (DCFTS-IMC) scheme; 2) a wordline-based serial access computing (WSAC) scheme; 3) an integration-based voltage-to-time converter (IVTC); and 4) a hidden-latency time-to-MAC value conversion (HLTMC) scheme. The proposed 22-nm 8-Mb resistive random access memory-CIM (ReRAM-CIM) macro was fabricated to demonstrate MAC operations with 8-b input, 8-b weight, and 19-b output. Our nvCIM macro achieved computing latency of 14.4 ns under 8-b precision with an energy efficiency of 21.6 TOPS/W.

Index Terms—Artificial intelligence (AI), computing-in-memory, convolutional neural network (CNN) edge processors, multiply-and-accumulate (MAC), resistive random access memory (ReRAM).

I. INTRODUCTION

ARTIFICIAL intelligence (AI) edge devices intended for convolutional neural networks (CNNs) require highly

parallel multiply-and-accumulate (MAC) operations to achieve high inference speeds and high energy efficiency. However, CNN operations generate a great deal of intermediate data, which is frequently moved between the processing element (PE) and memory in the conventional von Neumann-based computing architecture. This results in long latency and high energy consumption, particularly when applied to complex neural network models with high bit-precision. Embedded non-volatile computing-in-memory (nvCIM) [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20] is a promising approach to combine computing functions with non-volatile storage functions within the same macro to allow the parallel execution of multiple analog MAC operations in memory cell arrays. This makes it possible to reduce energy consumption and latency during system wake-up and computation.

Among the emerging non-volatile memory devices, one-transistor one-resistor (1T1R) single-level-cell (SLC) resistive random access memory (ReRAM) [26] is a feasible candidate for nvCIM, due to its production-ready manufacturability and high resistance ratio. Previous silicon verified ReRAM-based CIM macros have successfully demonstrated MAC operations from binary input and binary weight [3], [5], [7], [9] to 8-b input and 8-b weight [1], [2], [8]. Many of those works employ analog voltage on the bitline (BL) to generate memory cell current (I_{MC}) and accumulate I_{MC} on BL for MAC operations [1], [3], [4], [5], [6], [7], [8]; however, that approach increases energy consumption by necessitating the use of direct current in the memory cell array. Many nvCIM works employ high-resolution analog-to-digital converters (ADCs) to generate digital MAC values (MACVs) after an analog MAC signal is fully developed [1], [3], [4], [5], [6]; however, that approach increases computing latency when generating a high-precision digital output. Numerous works have reduced output precision in pursuit of shorter computing latency and lower energy consumption [1], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [20] for non-complex applications and datasets (e.g., MNIST [32]); however, that approach limits the output ratio (output precision/ideal output precision) and degrades inference accuracy when dealing with complex applications and/or datasets.

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AI-enabled edge devices with high inference accuracy require nvCIM with large memory capacity to accommodate complex neural network models with multiple-bit inputs and weights, while maintaining sufficient output ratio at the readout circuit. However, designing nvCIM circuits for MAC operations with multiple-bit input and multiple-bit weights imposes several challenges, including: 1) high energy consumption in the memory cell array due to the use of direct current on the BL during analog MAC operations; 2) long computing latency due to high parasitic load on the read path; and 3) long computing latency due to the separation of the analog computing phase from the ADC readout phase. In the current work, we developed a ReRAM-based nvCIM macro to overcome the abovementioned challenges using foundry-provided 22-nm ReRAM technology.

The remainder of this article is organized as follows. Section II outlines the challenges inherent in the design of multibit nvCIM and describes the macro structure of the proposed high-precision nvCIM macro. Section III describes the proposed direct-current free in-memory-computing technique used to generate analog MACV signals. Section IV describes the proposed hidden latency time-to-MACV readout scheme and corresponding readout circuit used to achieve high output precision and a high output ratio. Section V presents the simulated and measured performance results. Section VI concludes this work.

II. CHALLENGES IN MULTIBIT NVCIM AND THE STRUCTURE OF THE PROPOSED MULTIBIT NVCIM MACRO

A. Direct Current in nvCIM for MAC Computing

The use of conventional current-mode computing for MAC operations tends to induce multiple dc current branches in the memory cell array, as shown in Fig. 1(a). When read voltage is applied to the BL, the direct current flowing through the memory cell (I_{MC}) is in either high resistance state (I_{HRS}) or low resistance state (I_{LRS}). The accumulation of multiple I_{MC} branches on the BL increases the direct current on the BL (I_{BL}). The fact that nvCIM devices perform multiple MAC operations in parallel on different BLs increases the number of direct current branches. During MAC operations, dc current also appears in the peripheral readout circuits, including the amplifiers for the dynamic clamping of BL voltage, the circuit for current-to-voltage conversion in the analog domain, and current mode sense amplifiers.

Multiple dc-current branches can significantly increase energy consumption, peak current, and current density during the CNN inference stage. It can also lead to fluctuations in the power supply or VDD with a corresponding impact on readout accuracy.

B. Long Computing Latency in High-Precision MAC Operations

Most existing silicon verified nvCIM macros separate the analog MAC signal developing phase and ADC readout phase, as shown in Fig. 1(b). Most nvCIM macros are implemented in three operation phases: BL current developing phase, ADC phase, and digital shift-and-add (DSA) phase. During the

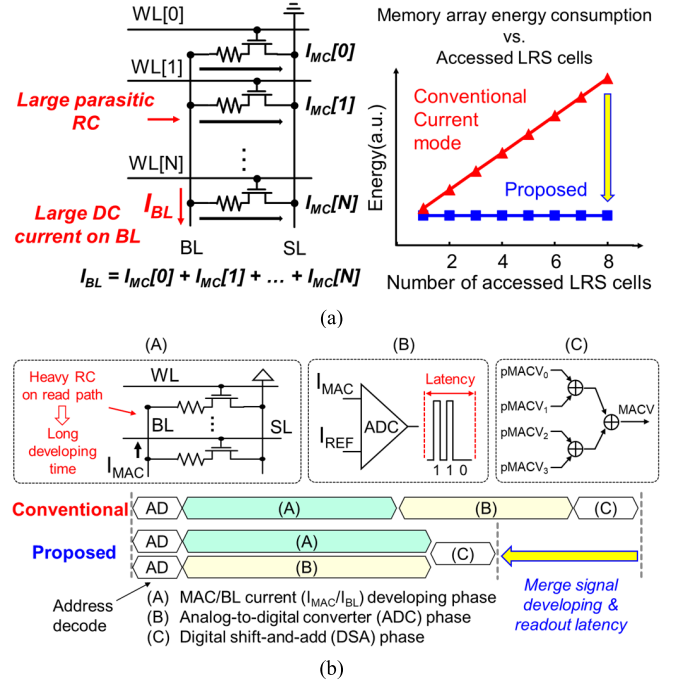


Fig. 1. Challenges in the development of nvCIMs. (a) Large energy consumption in the memory array. (b) Long computing latency for high-precision MAC operations.

BL-current developing phase, latency is the result of large parasitic RC load on the BL. Those macros with a higher memory capacity require a longer BL developing time, due to higher RC loading. Some previous works sought to amplify the BL current to a certain level for place-value computing by using current mirror [1], [4], [5] or switch to the voltage domain by using a transimpedance amplifier [1] to suppress the number of direct current branches. During the ADC phase, the ADC is toggled N times to generate an N -bit digital output. Under these conditions, nvCIMs with higher output precision suffer more from latency overhead. In the final phase, the DSA combines the ADC output [partial MACV (pMACV)] into a final multi-bit MAC output. In typical nvCIM works [1], [3], [4], [5], [6], overall computing latency is dominated by the sequential operation of an analog MAC signal developing phase followed by an ADC readout phase.

C. Structure of the Proposed High-Precision nvCIM Macro

To overcome the abovementioned challenges, we developed an 8-Mb ReRAM-based CIM macro with 8-b input (IN), 8-b weight (W), and 19-b output (OUT). The proposed scheme includes the following features: 1) direct-current-free time-space-based in-memory computing (DCFTS-IMC) to reduce direct current and energy consumption in the memory cell array; 2) a wordline (WL)-based serial access computing (WSAC) scheme to suppress energy consumption and peak current in the memory array and column-multiplexor (YMUX) during the inference stage; 3) an integration-based voltage-to-time converter (IVTC) to convert the dataline (DL) voltage-domain signal into a time-domain signal, while increasing the time-step signal margin to enhance readout accuracy; and

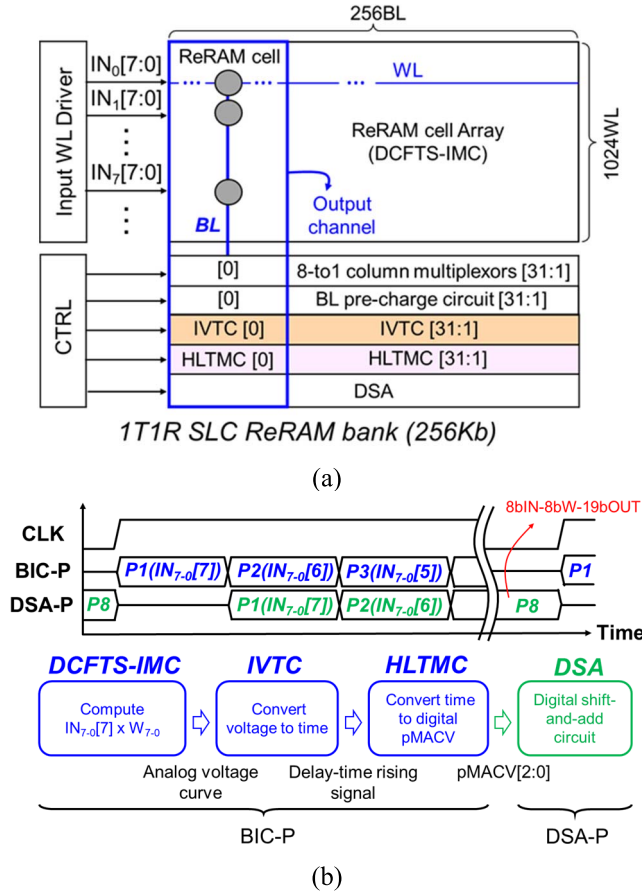


Fig. 2. (a) Macro-structure per memory bank. (b) Timing diagram of MAC operations in the proposed nvCIM macro.

4) a hidden-latency time-to-MAC value conversion (HLTMC) scheme to improve computing latency per output precision.

Fig. 2(a) illustrates the structure of the proposed 8-Mb ReRAM-based nvCIM macro, which includes 32 sub-banks. Each 256-kb memory bank performs MAC operations involving eight accumulations of 8-b input with 8-b weight ($IN_0[7:0]$ – $IN_7[7:0]$). Each memory bank also includes 32 output channels to support parallel MAC operations. The 8-b-weight data are stored in eight SLC ReRAM cells across eight columns on the same row (WL) in 2's-complement format. Each output channel comprises eight memory cells, one 8-to-1 YMUX, one BL pre-charge circuit, one IVTC, and one HLTMC. A single reconfigurable DSA circuit is shared by a memory bank.

Fig. 2(b) presents a timing diagram of the proposed macro. Each clock cycle includes a pipelined bitwise-input computing phase (BIC-P) and DSA phase (DSA-P) to complete N iterations for an N-bit input. For example, an 8-b input ($IN[7:0]$) requires no more than eight WL pulses across eight short BIC-Ps. The eight bitwise inputs in BIC-P1 are MSBs (i.e., $IN_0[7]$ – $IN_7[7]$), and the eight bitwise inputs in BIC-P2 are the second MSBs (i.e., $IN_0[6]$ – $IN_7[6]$). The last eight bitwise inputs in BIC-P8 are the LSBs (i.e., $IN_0[0]$ – $IN_7[0]$). The DCFTS-IMC, IVTC, and HLTMC perform in-memory MAC computation during the BIC-P phase, and the DSA composes the 3-b pMACVs (pMACV[2:0]) generated by HLTMC during

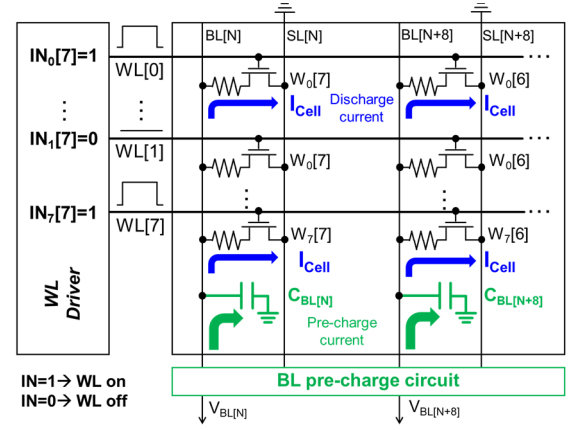


Fig. 3. Memory array structure and concept underlying DCFTS-IMC scheme.

the DSA phase. After eight DSA phases, the 19-b digital MACV (8-b IN–8-b W–8 accumulation) becomes available at the DSA output.

III. DIRECT-CURRENT-FREE TIME-SPACE-BASED MAC SIGNAL GENERATION SCHEME

A. Direct-Current-Free Time-Space-Based in-Memory Computing (DCFTS-IMC)

Fig. 3 illustrates the structure of the proposed memory cell array and the concept underlying the proposed DCFTS-IMC scheme. Each BIC-P can be divided into three sub-phases (SP-0–SP-2). During SP-0, all peripheral circuits are rapidly reset to their initial states. During SP-1, the selected BLs are pre-charged to target voltages (V_{BLP}), which do not exceed the read disturb voltage of the ReRAM devices. During SP-2, eight WLs (eight accumulations) are activated for the eight corresponding bitwise inputs. If the bitwise input is 0, then the corresponding WL is OFF. If the bitwise input is 1, then the corresponding WL is ON for a given period to discharge the BL capacitance (C_{BL}) via the accessed ReRAM cells. The discharged memory cell current (I_{Cell}) represents pMACV in the analog domain, which is the result of multiplying the WL (IN) by the cell data (W), as in the previous nvCIMs. The discharged BL current (I_{BL}) is the sum of eight I_{Cell} values for 1 b IN $\times$ 1 b W with eight accumulations.

The I_{BL} discharges BL parasitic capacitance (C_{BL}) at various discharge rates, and the BL voltage (V_{BL}) drops in accordance with pMACV. A larger I_{BL} indicates that a larger number of low-resistance cells (LRSs) have been accessed (i.e., higher pMACV) and the V_{BL} discharge rate is steeper. This reduces the BL discharge latency (t_{pMACV}) to meet the target BL voltage (V_{BL-TH}) for the voltage-to-time converter. A small I_{BL} indicates that a larger number of high-resistance cells (HRSs) have been accessed (i.e., lower pMACV) and V_{BL} discharge is more gradual. This increases the BL discharge latency (t_{pMACV}) to meet the target BL threshold voltage (V_{BL-TH}) for the voltage-to-time converter, as shown in Fig. 4(a).

Fig. 4(b) presents a truth table listing discharge latency as a function of pMACV. A larger pMACV indicates that more LRS memory cells have been accessed. Discharge latency

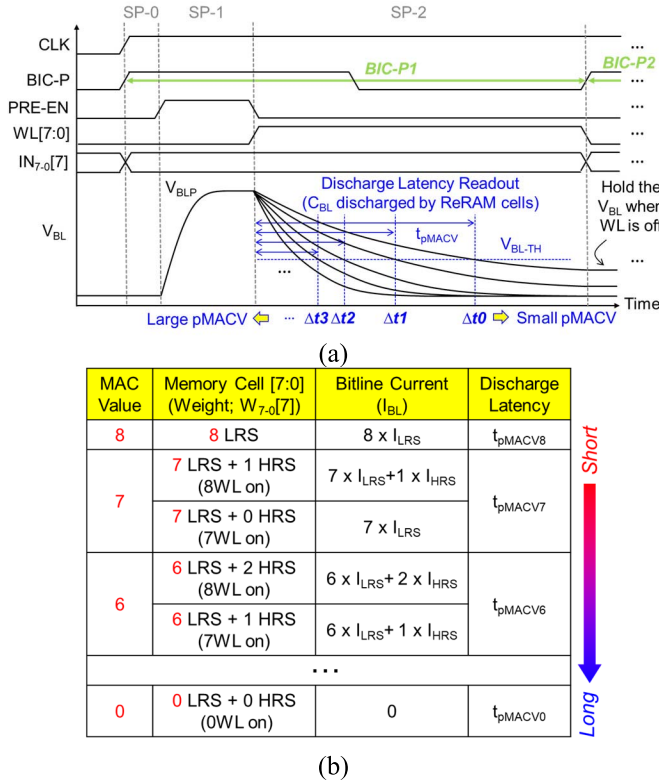


Fig. 4. (a) Timing waveform of the proposed DCFTS-IMC scheme. (b) Truth table of the proposed DCFTS-IMC.

decreases due to larger discharged I_{BL} . Note that the discharge latency for a given pMAC is presented as a range of values corresponding to various combinations of HRS cells and LRS cells. For example, pMACV = 7 may be associated with eight WLs switched ON (seven LRS cells plus one HRS cell). pMACV = 7 may also be associated with seven WLs switched ON (seven LRS cells plus zero HRS cells).

B. Wordline-Based Serial Access Computing (WSAC)

WSAC is meant to be implemented in conjunction with DCFTS-IMC to lower energy consumption in the memory cell array. WSAC reduces energy consumption by recycling the residual charge on the BL and reducing the toggling rate of the YMUX. Note that the YMUX and YMUX drivers employ high-voltage (2.5 V) devices to comply with write operations, albeit at the cost of energy consumption to switch the column address. Fig. 5 presents an operational flowchart and waveforms of the WSAC scheme.

The concept underlying WSAC involves the serial activation of different sets of WLs to access memory cells without toggling the column address. In each cycle, the DCFTS-IMC scheme activates eight WLs for in-memory MAC computation. Each memory bank includes 1024 WLs, such that the row address is 7 bit ($1024/8 = 128$; 7 bit). For instance, if in cycle 1, a set of eight WLs are activated for an 8-b IN-8-b W MAC operation using row address 7b'0000000 and column address Y_1 , then in cycle 2, another set of eight WLs will be activated for another 8-b IN-8-b W MAC operation using row address 7b'0000001 and column address Y_1 , as shown in Fig. 5. Generally, if in cycle K , a set of eight WLs are

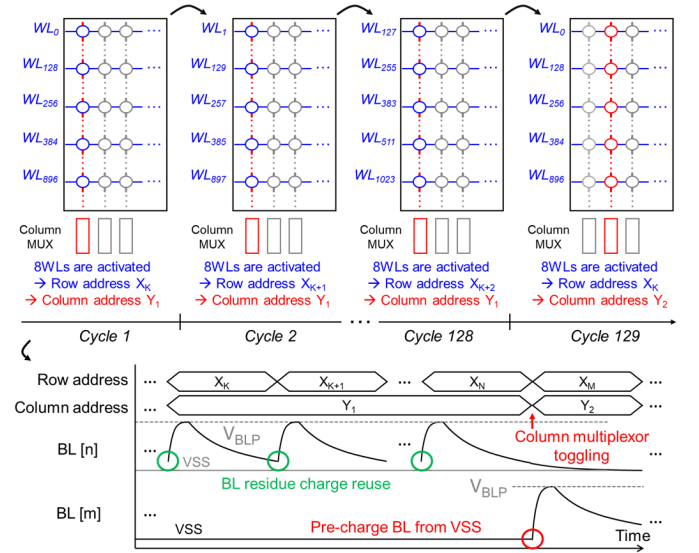


Fig. 5. Concept and computing flow of WSAC.

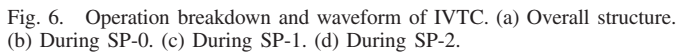
activated for an 8-b IN-8-b W MAC operation using row address X_K and column address Y_M , then in cycle $K + 1$, another set of eight WLs will be activated for another 8-b IN-8-b W MAC operation using row address X_{K+1} and column address Y_M . After the final set of WLs (row address = X_N) were accessed, the column address would switch to access other sets of BLs (e.g., Y_2) and enable other sets of eight WLs (row address = X_M). This would enable WSAC to access the memory cell array in an orderly manner and reuse the charges remaining on the BL. Note that these residual charges occur after performing DCFTS-IMC, thereby avoiding the need to charge the BL from 0 V to V_{BLP} before initiating each BIC-P phase. By reducing 2.5-V signal toggling in column-address drivers, this approach also makes it possible to reduce high-voltage energy consumption to below that of ReRAM macros in random access mode. The co-implementation of DCFTS-IMC with WSAC further alleviates peak current consumption by the memory cell array to improve the stability of the macro during the inference stage.

C. Integration-Based Voltage-to-Time Converter

The IVTC converts the discharge rate of DL voltage (V_{DL} ; equal to the V_{BL} of a selected column) into a delay-time rising signal. The IVTC also increases the time-step margin between the delay-time rising signals of neighboring pMACV pairs. As shown in Fig. 6(a), the IVTC comprises one coupling capacitor (C_0), one sampling capacitor (C_1), one integration capacitor (C_2), four initial transistors (N_0 , N_1 , N_3 , and P_1), one PMOS (P_0) as an integration current source, four switches (SW_1 – SW_4), one feedback transistor (P_2), one threshold-trigger NMOS (N_2), and one output inverter (INV_1).

The overall operation of IVTC can be divided into three sub-phases: circuit initialization during the first sub-phase (SP-0), V_{-P_0} sampling during the second sub-phase (SP-1), and integration with launch during the third sub-phase (SP-2). Note that the timing is synchronized with that of the DCFTS-IMC.

During SP-0 (circuit initialization phase), SW_1 is ON to reset node dc to the BL pre-charge voltage (V_{BLP}). N_0 , N_1 ,



During SP-2 (integration with launch phase), SW1 and SW3 are OFF, while SW2 and SW4 are ON. At the same time, the WL is ON to discharge parasitic capacitance (C_{BL}) on the selected BL and DLs via the ReRAM cells. The voltage drop on DL (ΔV_{DL} ; $V_{BLP} - V_{DL}$) is continuously ac

The 16-bit time-space code (pMACV-TC[15:0]) determined by TDC is latched by the 16 EDC cells. A larger pMACV makes it possible to generate a larger number of “1s” in the 16-bit time-space code. The 16-bit time-space code is then mapped using the entries in a TCT to generate a 3-b pMACV[2:0] output, as shown in Fig. 7(b). The HLTMC automatically generates TCT during macro boot-up using fixed input and ReRAM patterns pre-stored in the memory array. The TCT contains a counter and seven decision thresholds to calibrate pMACV-TC[15:0]. First, the TCT counts the number of 1s among pMACV-TC[15:0] using a counter and then sequentially compares the result with seven timing thresholds, as shown in Fig. 7(c). For example, the number of 1s in pMACV-TC[15:0] = 16'b0000000001111111 is 7. If in the decision branch, the number of 1s in a given pMACV-TC[15:0] is smaller than TH_N , then TCT maps pMACV-TC[15:0] to pMACV = N . If the number of 1s in a given pMACV-TC[15:0] is smaller than TH_6 , then TCT maps pMACV-TC[15:0] to pMACV = 6; otherwise, TCT maps pMACV-TC[15:0] to pMACV = 7. Note that the TCT is maintained by each HLTMC using respective seven timing thresholds (TH_0 – TH_6) to calibrate process and timing variation in each readout channel. Timing variations include near–far effects in the WL direction across multiple HLTMCs. Accordingly, the same

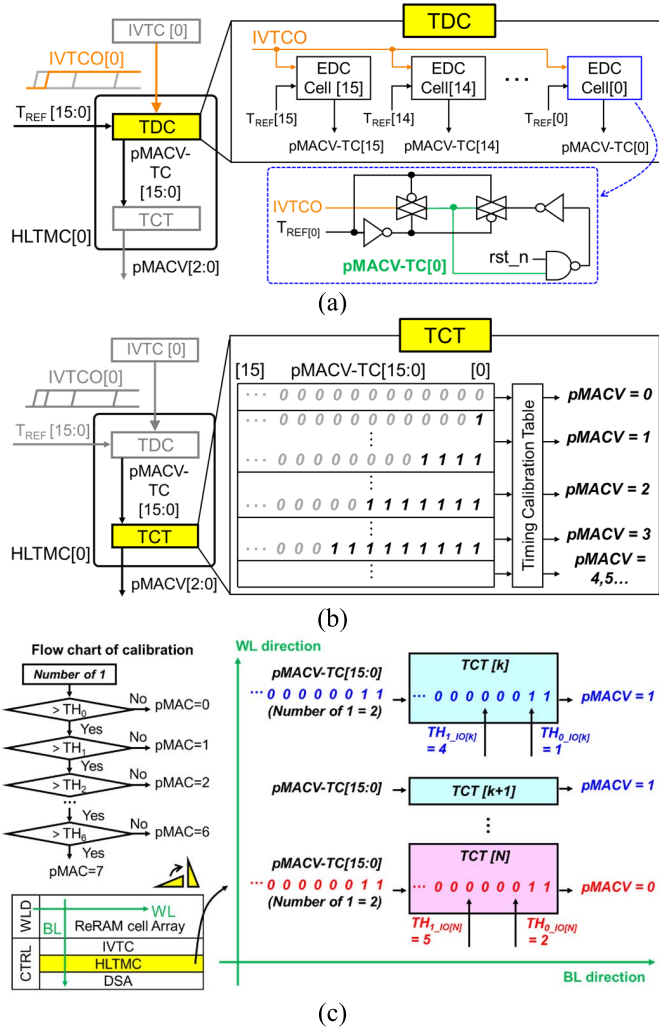


Fig. 7. Block diagram of HLTMC. (a) 16-bit TDC. (b) TCT. (c) Operation flow and example of calibration for pMACV using a TCT.

pMACV-TC[15:0] may be mapped to different pMACVs using different sets of thresholds (TH_0 – TH_6) under different output channels.

B. Digital Shift-and-Add Circuit (DSA)

The DSA circuit is a two-level structure. DSA level 1 (DSA-L1) combines pMACV[2:0] with the respective weight place value. DSA level 2 (DSA-L2) combines pMACV from DSA-L1 with the respective input place value.

DSA-L1 combines N pMACV[2:0] generated from HLTMC in N -bit weight mode. Take as an example the 8-bit weight, in which DSA-L1 combines eight sets of pMACV[2:0] (from HLTMC[0] to HLTMC[7]) to generate an 11-bit pMACV ($DSA_{L1}[10:0]$) value in each DSA-P, where $DSA_{L1}[10:0]$ refers to $IN_0[7:0] \cdot W_0[7:0] + IN_1[7:0] \cdot W_1[7:0] + \dots + IN_7[7:0] \cdot W_7[7:0]$. Here, pMACV₇[2:0] has the highest place value, whereas pMACV₀[2:0] has the lowest place value, as shown in Fig. 8(a).

DSA-L2 combines the M output generated from DSA_{L1} in M -bit input mode. Take as an example the 8-bit input mode, wherein DSA-L2 combines $DSA_{L1}[10:0]$ (from DSA-P1 to DSA-P8) to generate a 19-bit MACV ($DSA_{L2}[18:0]$),

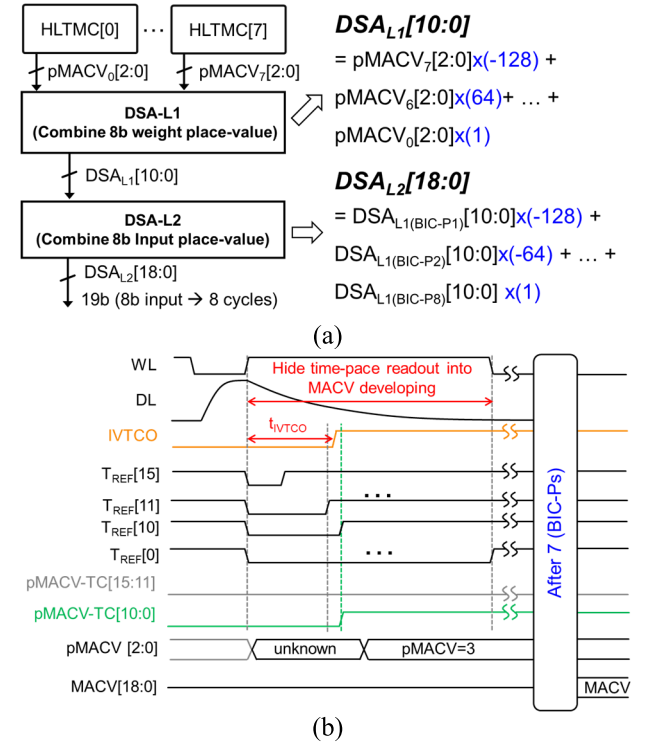


Fig. 8. (a) Structure of two-level DSA circuit and (b) overall waveform of HLTMC and DSA.

where $DSA_{L2}[18:0]$ refers to $IN_0[7:0] \cdot W_0[7:0] + IN_1[7:0] \cdot W_1[7:0] + \dots + IN_7[7:0] \cdot W_7[7:0]$. The overall waveform of HLTMC and DSA is shown in Fig. 8(b). The flexibility of the DSA circuit gives the proposed nvCIM macro multiple input–weight precision from binary input and binary weight to 8-bit input and 8-bit weight. The corresponding output precision ranges from 3 to 19 bit. Note that the binary input and binary weight should bypass the DSA, thereby directly outputting the results generated from HLTMC.

V. EXPERIMENTAL RESULTS AND PERFORMANCE OF THE PROPOSED SCHEMES

A. Performance of the Proposed Schemes

Fig. 9 presents the simulated energy consumption in a memory cell array with 1024 rows. The analysis considered the probable number of accessed LRS cells and active WL during the inference stage of the ResNet-20 model when applied to the CIFAR-100 dataset. When using a conventional current-mode computing scheme, the increase in BL current is proportional to the number of WLs that are turned on. A larger MACV will also induce larger BL current due to a larger number of LRS cells being accessed. However, under the proposed DCFTS-IMC scheme, energy consumption in memory cell array does not depend on the number of activated WLs or LRS cells. Rather, energy consumption is dominated by parasitic capacitance on the BL and BL pre-charged voltage (V_{BLP}). Eliminating the use of direct-current during the MAC computing phase was shown to reduce energy consumption in memory cell array by $2.06\times$ to $16.5\times$, compared to that of the conventional current-mode computing scheme. Overall, the

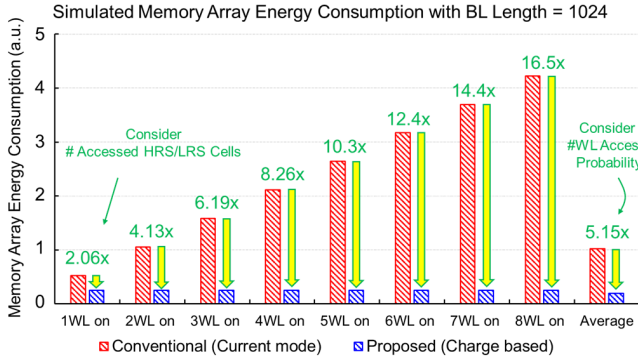


Fig. 9. Simulated energy consumption in memory cell arrays based on conventional current-mode computing methodology and the proposed DCFTS-IMC computing methodology.

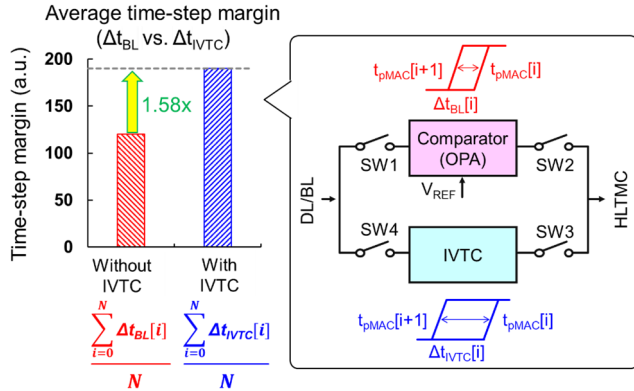


Fig. 10. Simulated time-step margin with and without IVTC.

energy consumption of the proposed scheme is 5.15 $\times$ smaller than that of the conventional approach.

Fig. 10 presents the simulated time-step margin with and without the proposed IVTC circuit. The time-step margin at node IVTCO (Δt_{IVTC}) is increased in 1.58 $\times$ compared to that without using IVTC (Δt_{BL}). The scheme without IVTC uses a comparator (operational amplifier) with an adjustable reference voltage. When discharge voltage on the BL/DL falls below the reference voltage, the comparator generates a rising signal. To ensure a fair comparison with the IVTC scheme, we swept the reference voltage in order to maximize the signal margin. Note that in the comparison of time-step margins in Fig. 10, we averaged the time-step margins between each peripheral pMACV. Fig. 11 shows the comparison of the computing latency per output bit in the proposed scheme with that of previous works [34], [35], [36], [37]. Note that the existing silicon verified nvCIM works vary in terms of output precision and computing latency; therefore, we normalized these indexes to latency per output bit to enable a fair comparison. The parallel operation involving an analog MAC signal developing phase and readout MAC-signal phase increased the latency per output bit by 1.362 to 8.30 $\times$, compared to previous works.

Fig. 12 shows the comparison of the simulated macro-level energy consumption with that of previous work [1], [4]. Note that we normalized the energy consumption to 8-b input and 8-b weight precision per MAC operation. The readout circuit included peripheral circuits for computing in the analog domain, one ADC, one reference generator, and one DSA

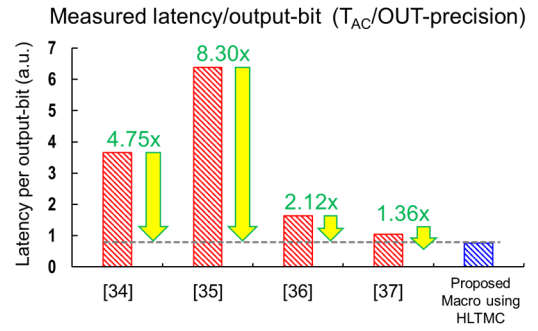


Fig. 11. Measured computing latency per output bit compared with other recent silicon verified nvCIM works.

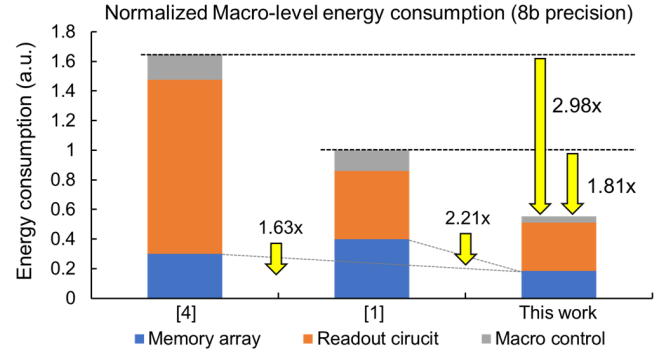


Fig. 12. Comparison of macro-level energy consumption in the proposed scheme and previous works.

circuit. Macro control involved main controller, WL drivers, and YMUX. The proposed schemes were shown to reduce macro-level energy consumption by 2.98 $\times$, compared to that of [4] using a current-mode operations for in-memory computing and analog readout circuitry. Macro-level energy consumption in this work was 1.81 $\times$ lower than that of [1] using current-mode in-memory computing in conjunction with voltage-mode analog readout circuitry.

B. Measurement Results

We evaluated the efficacy of the proposed schemes by fabricating a test chip that included an 8-Mb nvCIM macro fabricated using 22-nm CMOS logic process in conjunction with foundry-provided logic-compatible 1T1R SLC ReRAM devices. Fig. 13 presents a die photograph of the chip and a test chip summary table for MAC operations from binary to 8-b precision.

Fig. 14 presents a breakdown of the measured energy consumption in the proposed 8-Mb ReRAM nvCIM macro, based on the split power pins implemented in the test chip. Overall, the energy consumption of memory cell array, which involved in pre-charging the parasitic capacitance on BL, accounted for 31.47% of the total energy consumed by the 8-Mb nvCIM macro. HLTMC accounted for 31.11% and IVTC accounted for 28.82%. Core VDD (including the WL drivers, timing generation circuits, control circuits, and buffers) accounted for 6.7%. High VDD (including YMUXs, high-voltage drivers, and programming circuits) accounted for 1.9%.

Fig. 15 presents an area breakdown of the proposed 8-Mb ReRAM nvCIM macro. The memory cell array accounted for

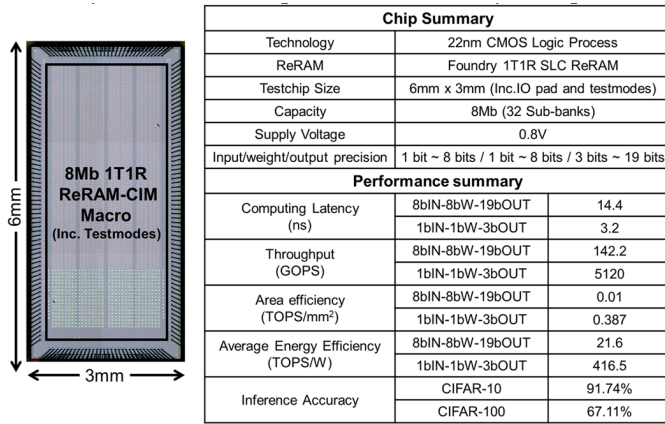


Fig. 13. Die photograph and summary table.

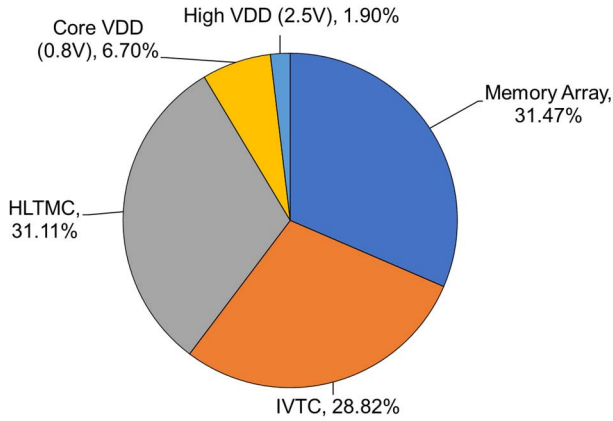


Fig. 14. Energy breakdown of the proposed ReRAM-based nvCIM macro.

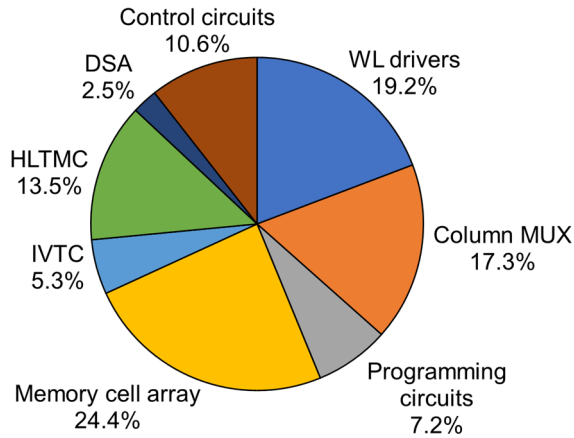


Fig. 15. Area breakdown of the proposed ReRAM-based nvCIM macro.

24.4% of the total area with the WL driver accounting for 19.2% and the column mux (high-voltage devices) accounting for 17.3%. The readout circuit (including the IVTC, HLTMC, and DSA) accounted for 21.3%. The control circuits (including timing control, column decoder, row decoder, mode control, and buffers) accounted for 10.6%. The programming circuits accounted for 7.2%.

Fig. 16 presents the computing latency (T_{AC}) for each BIC-P, and the test-mode circuit is used to measure computing latency. Note that in the test chip, we employed the DFF-based path-delay exclusion scheme [5] to enable the extraction of MAC computing latency. The captured waveform confirmed

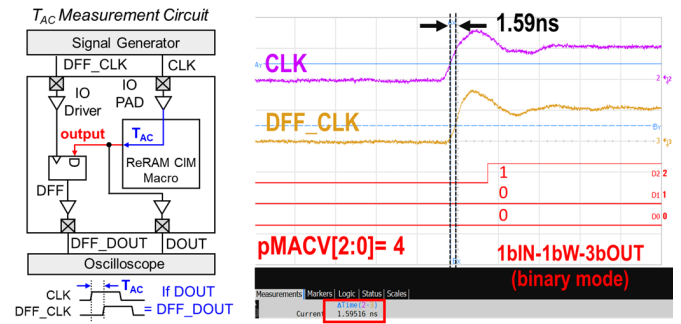
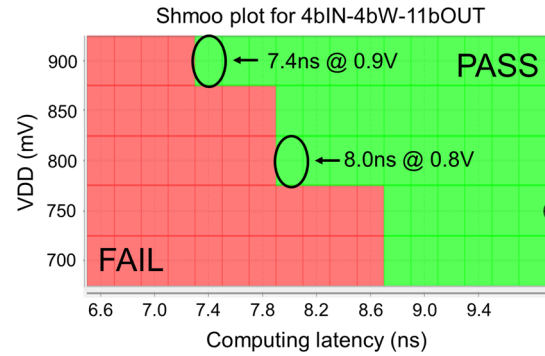
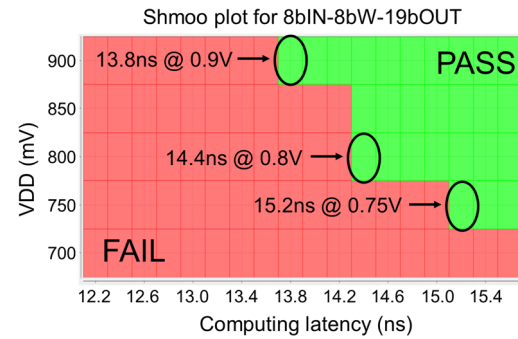


Fig. 16. Measured computing latency in BIC-P phase using the DFF-based path-delay exclusion scheme.



(a)



(b)

Fig. 17. Measured Shmoo plot of VDD versus computing latency using two input-weight configurations. (a) 4-b input, 4-b weight, and 11-b output. (b) 8-b input, 8-b weight, and 19-b output.

that the computing latency was roughly 1.6 ns for each BIC-P, and the example output of pMACV (pMACV[2:0]) is 4.

Fig. 17(a) presents the measured Shmoo plot obtained from the 8-Mb ReRAM-CIM macro for MAC operations involving eight accumulations of 4-b input with 4-b weight (4-b IN $\times$ 4-b W) and 11-b output. The measured MAC-computing latency (T_{MAC}) was 8 ns at VDD = 0.8 V and 7.4 ns at VDD = 0.9 V. When performing MAC operations with eight accumulations of 8-b input with 8-b weight (8-b IN $\times$ 8-b W) and 19-b output, the ReRAM-CIM macro achieved T_{MAC} = 14.4 ns at VDD = 0.8 V and T_{MAC} = 13.8 ns at VDD = 0.9 V, as shown in Fig. 17(b).

Fig. 18(a) presents the measured readout distribution (pMACV[2:0]) of a readout channel versus the ideal MACVs. Note that the readout distributions were measured under real-world variations in ReRAM cell resistance and process

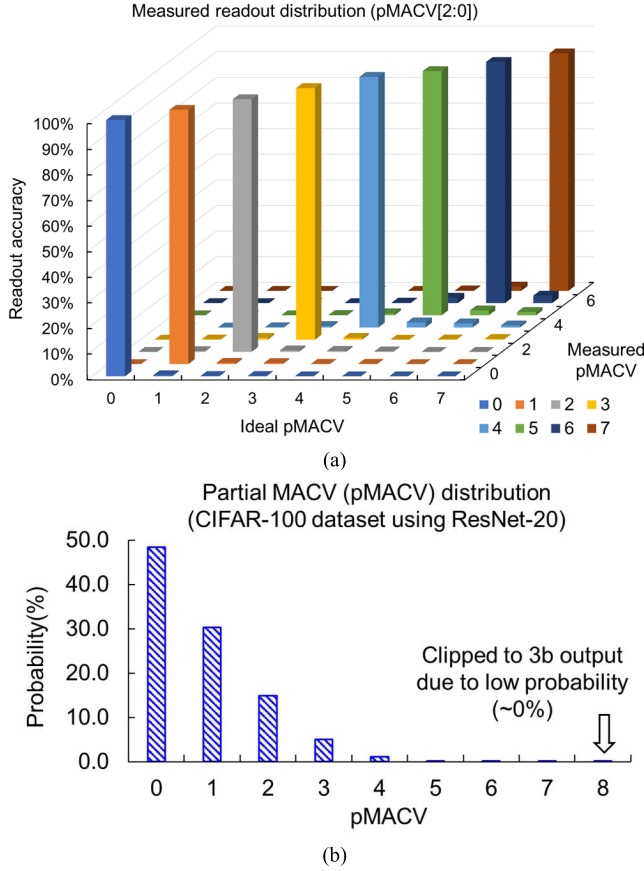


Fig. 18. (a) Measured readout distribution (pMACV[2:0]) of readout channel. (b) Simulated pMACV distribution in classifying images in the CIFAR-100 dataset using the ResNet-20 model.

variation in transistors on the readout path. The measured readout yield ranged from 96.2% to 100% for lower pMACVs (0–4), which frequently occur in neural network models. The readout yield ranged from 92.70% to 95.3% for higher pMACVs (5–7), which occur less frequently. Fig. 18(b) presents a simulated pMACV distribution in the proposed nvCIM macro when used to classify 10 000 test images from the CIFAR-100 dataset using a ResNet-20 model.

Fig. 19 presents the readout accuracy of an output channel without calibration or with calibration using exclusive timing thresholds. Readout accuracy in the case without calibration was measured using default thresholds values from output channel [0]. Readout accuracy in the case without calibration was between 3.4% and 18.6% lower.

A higher V_{BLP} would preserve the signal-to-noise ratio (SNR); however, it would result in longer precharge latency and higher energy consumption. In the experiment, V_{BLP} was set at 0.3 V, which is the maximum voltage without read disturbance. In the current design, readout accuracy was prioritized over latency and energy consumption. Fig. 20 presents a comparison of energy efficiency, throughput and weighted readout accuracy under various V_{BLP} values (0.3, 0.2, and 0.1 V) in 8-b IN–8-b W–19-b OUT mode. Note that weighted readout accuracy was computed by summing up the readout accuracy of each pMACV, the result of which was multiplied with the probability of its appearing

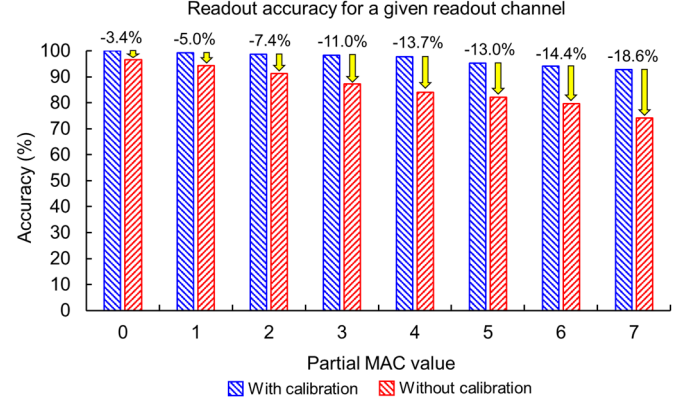


Fig. 19. Readout accuracy of pMACVs with or without TCT calibration.

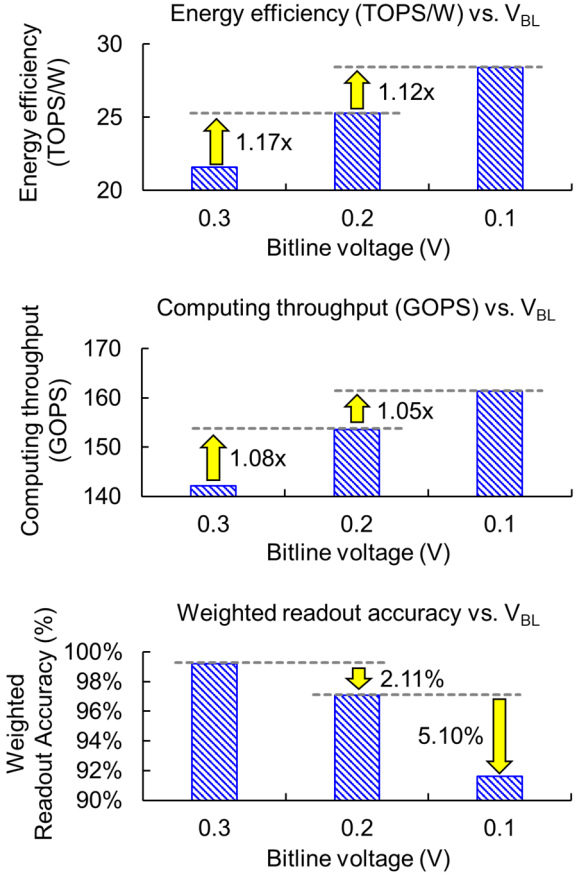


Fig. 20. Design tradeoff between energy efficiency, computing throughput, readout accuracy, and BL precharge voltage.

during the inference stage, as follows.

$$\text{Weighted readout accuracy} = \sum_{pMACV=0}^{pMACV=7} \text{Probability}(pMACV) \times \text{Yield}(pMACV).$$

Regulating BL voltage from 0.3 to 0.1 V increased energy efficiency from 21.6 to 28.4 TOPS/W by reducing energy consumption in the memory array. This also led to an increase in computing throughput from 142.2 to 161.4 GOPS, due to a decrease in precharge latency. However, this led to a reduction in weighted readout accuracy from 99.2% to 91.6%, due to a decrease in the SNR.

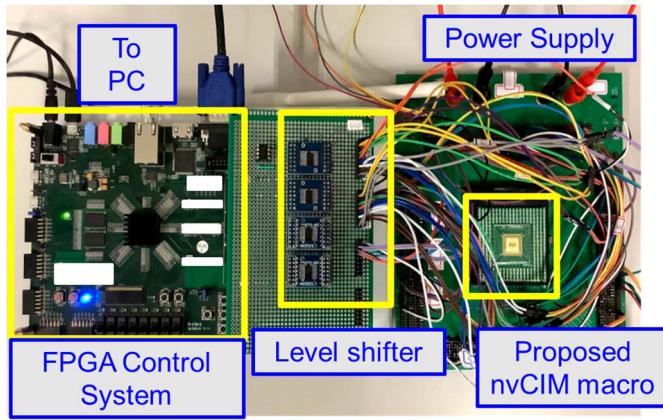


Fig. 21. Demonstration system for convolutional neural network with FPGA host.

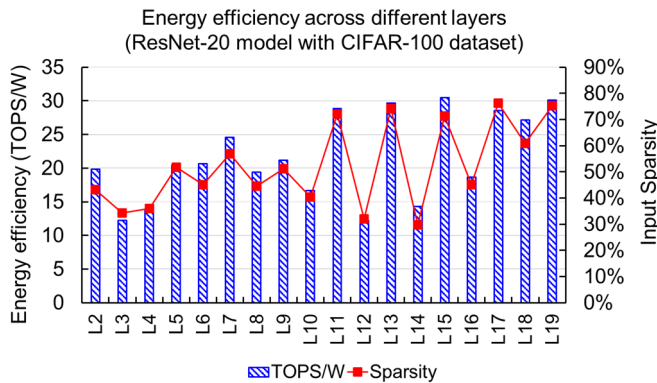


Fig. 22. Energy efficiency of the proposed ReRAM-based nvCIM macro in 8-b precision mode across various layers when applying the ResNet-20 model to the CIFAR-100 dataset.

Fig. 21 illustrates the setup of demonstration system involving the application of the Resnet-20 model to the CIFAR-100 dataset. The demonstration system included a test chip with the 8-Mb ReRAM-CIM macro and a field-programmable gate array (FPGA) host with on-board level shifters. Note that the level shifters were used to bridge the various voltage domains involving interchip signals between the ReRAM-CIM macro and the IO pins of the FPGA board. The ReRAM-CIM macro was used for MAC operations, and the FPGA board was used to control dataflow, activation, batch normalization, and channel accumulation in the neural network.

Fig. 22 presents the energy efficiency of the proposed ReRAM-based nvCIM macro in 8-b precision mode across various layers and wordwise sparsity levels when applying the ResNet-20 neural network model to the CIFAR-100 dataset. Overall, the energy efficiency was proportional to input sparsity. In layer 15 (L15), peak energy efficiency was achieved under input sparsity of 71.25% and the average energy efficiency was 21.6 TOPS/W.

As shown in Fig. 23, when applied to the CIFAR-10 dataset in 8-b IN–8-b W–19 b OUT mode, the degradation in inference accuracy varied with the neural network model as follows: ResNet-50 (0.79%), VGG-16 (0.63%), and ResNet-20 (0.46%). When applied to the CIFAR-10 dataset in 8-b IN–8-b W–19-b OUT mode, the degradation in inference accuracy was as follows: ResNet-50 (1.11%), VGG-16 (1.1%),

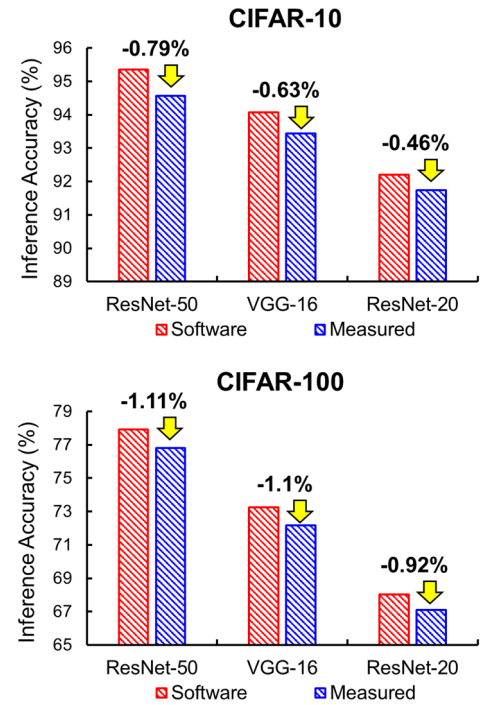


Fig. 23. Inference accuracy degradation as a function of neural network model (ResNet-50, VGG-16, and ResNet-20) when applied to the CIFAR-100 and CIFAR-10 datasets.

	This work	ISSCC2021 [37]		ISSCC2021 [6]	ISSCC2020 [36]		ISSCC2020 [35]	ISSCC2019 [34]	
Technology (nm)	22	22		40	22		130	55	
Capacity	8Mb	4Mb		64Kb	2Mb		158.8Kb	1Mb	
Supply voltage (volt)	0.8	0.8		0.9	0.8		4.2	1	
Input precision (bit)	1-8	1	4	8	1-8	1	4	1	2
Weight precision (bit)	1-8	2	4	8	1-8	2	4	2	3
Output precision (bit)	3-19	4	10	14	4 - 20	6	11	8	4
Output ratio	1	1	1	0.78	1	1	0.92	0.67	0.57
CIM mode	Voltage	Current		Current	Current		Current	Current	
Readout circuit	Time	Voltage		Voltage	Current		Current	Current	
Throughput (GOPS)	5120 - 142.2	417.96	99.42	34.59	N/A	26.12	13.99	3068.49	12.26
Computing latency (ns)	3.2 - 14.4	4.90	10.30	14.80	N/A	9.80	18.30	51.1	11.75
Area efficiency (TOPS/mm ²)	0.387-0.01	0.081	0.019	0.007	N/A	0.005	0.003	0.14	0.002
Energy efficiency (TOPS/W)	416.5 - 21.6	195.70	47.26	11.91	56.67 - 4.15	121.38	28.93	79.06	53.17
FoM	416.5 - 1382.4	391.4	756.2	594.5	265.6	242.8	425.8	106	60.6

FoM = Energy efficiency × Input precision × Weight precision × Output ratio

Fig. 24. Comparison table.

and ResNet-20 (0.92%). Note that testing was performed on 10 000 images, respectively, from the CIFAR-100 and CIFAR-10 datasets.

Fig. 24 presents a comparison table of the proposed 8-Mb nvCIM macro and previous silicon verified nvCIM works. Note that the proposed scheme had the highest ReRAM capacity to accommodate complex neural network models. The proposed scheme also used voltage-domain computing in conjunction with time-domain readout to enhance energy efficiency. Previous works using current-mode computing and readout operations suffered in terms of energy efficiency. The proposed nvCIM macro achieved the highest figure of merit (FoM) across binary to 8-bit input–weight precision configurations, where FoM was calculated as energy efficiency × input precision × weight-precision × output ratio.

VI. CONCLUSION

This article reports an 8-Mb ReRAM-based nvCIM CMOS-integrated macro capable MAC operations with 8-bit precision, while providing high energy efficiency, short latency, and inference accuracy sufficient for AI edge devices. The proposed macro features DCFTS-IMC and WSAC schemes to reduce the energy consumption by the memory array as well as an IVTC and HLTMC scheme to reduce computing latency and increase readout accuracy. The efficacy of the proposed schemes enabled the proposed ReRAM-based CIM macro to achieve short computing latency (3.2–14.4 ns) with high energy efficiency (21.6–416.5 TOPS/W), while performing MAC operations with binary to 8-b precision.

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